DATA ANALYTICS INTERNSHIP

The internship report submitted in partial fulfilment of the requirements for the award of degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JNTUA COLLEGE OF ENGINEERING (Autonomous), PULIVENDULA

ANDHRA PRADESH - INDIA

JNTUA COLLEGE OF ENGINEERING, PULIVENDULA (Autonomous)

ANDHRA PRADESH – INDIA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify the internship report on **Data Analytics Internship** is a Bonafide record of Internship work done and submitted by **B.Sravanthi** bearing Roll No.**22191A0536**,in partial fulfilment for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

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ACKNOWLEDGEMENT

It is a great pleasure to express a deep sense of gratitude and veneration to the internship provider **Digit it** for giving us an internship opportunity to undergo an internship in **Data analytics internship**.

I express my sincere thanks to **Dr. D. Vishnu Vardhan,** Principal of JNTUA College of Engineering (Autonomous), Pulivendula, for his encouragement and for providing the opportunity for the students to go and work in the field with the advent of internship while pursuing B.Tech IVth year.

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Foremost, we gratefully express our sincere gratitude to our respected **Dr. S. Jessica Saritha,** Assistant Professor, Head of the Department, Department of Computer Science and Engineering, JNTUA College of Engineering, Pulivendula for her valuable guidance, timely suggestions, and lively intense throughout the work. We are always thankful to her for providing us with the opportunity to do work.

The real mentor and motivator of this project **Dr.K.BalaChandra Reddy**, Assistant Professor of Computer Science and Engineering Department, JNTUA College of Engineering, Pulivendula. His wide knowledge and logical way of thinking have made a deep impression on us. His understanding, encouragement and personal guidance have provided the basis for this Internship. His source of inspiration for innovative ideas and his kind support is well to all his students and colleagues.

With gratitude,

B.Sravanthi

(22191A0536)

Internship Weekly Report

Week-1

(From 19-01-2026 to 24-01-2026)

Objective of the Activity Done:

To understand the internship workflow, tools, and overall data analytics process.

Day	Activities	Learning Outcomes
Day 1	Introduction to internship structure and understanding the role of data analysis in real-world applications.	Understood the workflow and importance of data analytics in decision-making.
Day 2	Explored project requirements and dataset related to food trends.	Gained clarity on dataset structure and project scope.
Day 3	Studied dataset attributes such as categories, preferences, and customer behavior.	Learned how different data fields contribute to analysis.
Day 4	Performed initial data inspection using Excel.	Identified missing values and inconsistencies in data.
Day 5	Observed data patterns and irregularities.	Understood the need for preprocessing before analysis.
Day 6	Planned approach for data cleaning and analysis workflow.	Learned structured approach to data analytics

Week-2

(From 26-01-2025 to 31-01-2025)

Objective of the Activity Done:
To perform data cleaning and preprocessing.

Day	Activity / Tasks Done	Learning Outcome
Day 1	Identified null values and irrelevant fields.	Learned techniques for detecting incomplete data.
Day 2	Removed duplicates and handled missing values.	Improved data cleaning skills.
Day 3	Standardized formats across dataset.	Understood importance of consistency.
Day 4	Verified cleaned data using Excel.	Ensured data accuracy.
Day 5	Structured dataset for analysis.	Prepared data for further processing.
Day 6	Reviewed cleaned dataset.	Strengthened preprocessing understanding.

Week-3

(From 02-02-2026 to 07-02-2026)

Objective of the Activity Done:

To begin data analysis using SQL and analytical techniques.

Day	Activities	Learning Outcomes
Day 1	Used SQL to extract relevant data.	Learned query-based data retrieval.
Day 2	Filtered data based on categories	Improved segmentation skills.
Day 3	Analyzed customer preferences.	Gained insight into behavior patterns.
Day 4	Performed basic analytical operations.	Developed analytical thinking.
Day 5	Grouped and summarized data.	Learned aggregation techniques.
Day 6	Documented findings.	Improved reporting skills.

Week-4

(From 09-02-2025 to 14-02-2026)

Objective of the Activity Done:

To create visualizations using Power BI

Day	Activities	Learning Outcomes
Day 1	Imported dataset into Power BI.	Learned data integration.
Day 2	Created basic charts.	Understood visualization techniques.
Day 3	Developed visuals for trends.	Learned to present insights.
Day 4	Improved layout and formatting.	Enhanced clarity in visuals.
Day 5	Added filters and slicers.	Improved interactivity.
Day 6	Refined visualizations.	Improved accuracy and present
Extra	Doubt session and grand quiz.	Clarified queries and reinforced overall understanding.

Week-5

(From 16-02-2026 to 21-02-2026)

Objective of the Activity Done:
To initiate project work on food trends analysis

Day	Activities	Learning Outcomes
Day 1	Defined project scope and metrics.	Learned project planning.
Day 2	Explored dataset deeply.	Improved understanding of attributes.
Day 3	Created initial visual insights.	Learned data-to-visual conversion.
Day 4	Analyzed ordering patterns.	Gained deeper insights.
Day 5	Improved visual outputs.	Enhanced presentation skills.
Day 6	Reviewed progress.	Learned iterative improvement.

Week-6

(From 23-02-2026 to 28-02-2026)

Objective of the Activity Done:

To develop dashboards and improve analysis

Day	Activity / Tasks Done	Learning Outcome
Day 1	Designed dashboard layout.	Learned structuring dashboards.
Day 2	Integrated multiple visuals.	Improved data storytelling.
Day 3	Added interactivity features.	Enhanced usability.
Day 4	Refined visual elements.	Improved clarity.
Day 5	Tested dashboards.	Learned debugging.
Day 6	Finalized dashboard.	Gained confidence.

Week-7

(From 02-03-2026 to 07-03-2026)

Objective of the Activity Done:

To complete initial project version.

Day	Activities	Learning Outcomes
Day 1	Reviewed dashboards.	Improved quality.
Day 2	Made corrections.	Learned refinement.
Day 3	Finalized initial version.	Strengthened output.
Day 4	Prepared documentation.	Improved reporting.
Day 5	Reviewed insights.	Improved interpretation.
Day 6	Completed first phase.	Gained project clarity.

Week-8

(From 09-03-2026 to 14-03-2026)

Objectives of the Activity Done:

To refine the existing analysis, improve data accuracy, and enhance insights by revisiting the dataset and correcting inconsistencies.

Day	Activities	Learning Outcomes
Day 1	Revisited the dataset and reviewed previously performed analysis to identify any inconsistencies.	Understood the importance of revisiting data to improve accuracy of results.
Day 2	Modified and optimized SQL queries for better data extraction and filtering	Improved query writing and data retrieval efficiency.
Day 3	Re-analyzed customer preferences and food category trends using refined data.	Gained deeper insights into customer behavior patterns.
Day 4	Corrected errors in earlier analysis and updated calculated results.	Learned the importance of iterative improvement in data analysis.
Day 5	Compared different food categories and trends for better understanding.	Developed comparative analysis skills.
Day 6	Documented updated findings and refined insights.	Improved documentation and reporting clarity.

Week-9

(From 16-03-2026 to 21-03-2026)

Objective of the Activity Done:

To enhance the quality of visualizations and improve dashboard clarity for better interpretation of insights.

Day	Activities	Learning Outcomes
Day 1	Reviewed existing dashboards and identified areas for improvement.	Learned to evaluate visualization effectiveness.
Day 2	Modified charts to improve clarity and readability.	Improved data presentation skills.
Day 3	Enhanced layout and arrangement of dashboard components.	Learned structured dashboard design.
Day 4	Added filters and slicers to improve interactivity.	Understood importance of user interaction.
Day 5	Tested dashboard functionality with different inputs.	Improved usability testing skills.
Day 6	Finalized updated dashboards with improved visuals.	Strengthened visualization techniques.

Week-10

(From 23-03-2026 to 28-03-2026)

Objectives of the Activity Done:

To understand version control concepts and the role of Git in managing project files and tracking changes.

Day	Activities	Learning Outcomes
Day 1	Introduction to version control systems and importance of Git in project management.	Understood the need for version control in real-world projects.
Day 2	Installed Git and configured basic settings.	Learned initial setup of Git environment.
Day 3	Practiced basic Git commands such as init, add, and commit.	Understood how to track changes in files.
Day 4	Created local repositories and managed project files.	Learned repository management concepts.
Day 5	Worked on tracking changes and maintaining versions of files.	Gained knowledge of version history.
Day 6	Reviewed Git workflow and practiced commands.	Strengthened understanding of Git basics.

Week-11

(From 30-03-2026 to 04-04-2026)

Objectives of the Activity Done:

To understand GitHub and manage remote repositories for project collaboration and storage.

Day	Activities	Learning Outcomes
Day 1	Introduction to GitHub and its features.	Understood role of GitHub in project hosting.
Day 2	Created GitHub account and repository.	Learned repository creation process.
Day 3	Connected local repository to GitHub.	Understood remote repository linkage.
Day 4	Pushed project files to GitHub repository.	Learned how to upload code and files.
Day 5	Updated project files and committed changes.	Improved version tracking skills.
Day 6	Explored repository structure and file management.	Learned organization of project files.

Week-12

(From 06-04-2026 to 11-04-2026)

Objectives of the Activity Done:

To manage project updates and maintain version control effectively using Git and GitHub.

Day	Activities	Learning Outcomes
Day 1	Made updates to project files and tracked changes.	Learned updating project versions.
Day 2	Practiced commit messages for better tracking.	Understood importance of meaningful commits.
Day 3	Worked on multiple file updates in repository.	Improved file management skills.
Day 4	Pulled latest changes from repository.	Learned synchronization between local and remote.
Day 5	Resolved minor issues during file updates.	Developed problem-solving skills.
Day 6	Reviewed project versions and changes.	Strengthened understanding of version history.

Week-13

(From 13-04-2026 to 18-04-2026)

Objectives of the Activity Done:

To integrate project work with GitHub and maintain proper documentation and version control.

Day	Activities	Learning Outcomes
Day 1	Uploaded updated project files and dashboards.	Learned project integration with GitHub.
Day 2	Organized repository files and folders.	Improved project structuring skills.
Day 3	Added project documentation to repository.	Understood importance of documentation.
Day 4	Reviewed repository for completeness.	Learned quality checking.
Day 5	Updated files based on feedback.	Improved iterative development.
Day 6	Finalized repository structure.	Strengthened project management skills.

Week-14

(From 20-04-2026 to 25-04-2026)

Objectives of the Activity Done:

To finalize project submission and ensure proper version control and documentation.

Day	Activities	Learning Outcomes
Day 1	Reviewed complete project in repository.	Ensured project completeness.
Day 2	Verified all files and documentation.	Improved attention to detail.
Day 3	Made final updates and commits.	Learned final version control practices.
Day 4	Prepared project for final submission.	Understood submission readiness.
Day 5	Submitted project through GitHub.	Learned final deployment process.
Day 6	Conducted final review and closure.	Gained overall project experience.

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ABSTRACT

This internship report presents the work carried out during my internship in the field of Data Analytics at Digit IT, Hyderabad. The project titled "*Facebook Food Trends Analysis: Understanding Customer Preferences*" focuses on analyzing customer behavior and identifying patterns in food ordering data.

The main objective of this internship was to gain practical exposure to the complete data analytics workflow, including data collection, cleaning, transformation, analysis, and visualization. Tools and technologies such as SQL, Python, Microsoft Excel, and Power BI were used throughout the project.

The process began with understanding the dataset and extracting relevant information using SQL. The data was then cleaned and processed to remove inconsistencies and prepare it for analysis. Further analysis was performed using Python and Excel to identify trends and patterns. Finally, interactive dashboards were created in Power BI to represent insights in a clear and meaningful way.

The internship provided valuable hands-on experience in transforming raw data into actionable insights. The final dashboards help in understanding customer preferences, popular food categories, and ordering patterns, which can support business decision-making.

This project enhanced my technical skills, analytical thinking, and ability to present data effectively, preparing me for real-world data analytics applications.

INTRODUCTION

Project Overview:

In today's competitive food industry, understanding customer preferences plays a crucial role in improving services and increasing business growth. With large amounts of data generated from customer orders, it becomes difficult to manually identify meaningful patterns and trends.

This project focuses on analyzing food ordering data to understand customer preferences and trends using data analytics techniques. The dataset includes attributes such as food categories, order frequency, customer ratings, and preferences. The aim is to extract useful insights that can help businesses make better decisions.

By applying data analysis and visualization techniques, raw data is transformed into meaningful insights through interactive dashboards. These insights help in identifying popular food items, customer behavior, and overall trends in the food industry.

Purpose:

The main **purpose** of this project is to analyze customer food preferences and provide clear insights that support decision-making.

Through this project, the following objectives are achieved:

- Analyze customer ordering behavior and preferences
- Identify popular food categories and trends
- Convert raw data into meaningful visual insights
- Create interactive dashboards for better understanding
- Improve analytical and problem-solving skills

ABOUT THE INTERNSHIP

The internship at Digit IT, Hyderabad was designed to provide practical exposure to real-world data analytics tasks. During this internship, I worked as a Data Analyst Intern and gained hands-on experience in analyzing datasets and generating meaningful insights.

The internship focused on applying data analytics techniques to understand and solve business-related problems. I worked on the project titled "**Facebook Food Trends Analysis: Understanding Customer Preferences**", where the goal was to analyze food-related data and identify patterns in customer behavior.

Throughout the internship, I was involved in:

- Understanding and exploring datasets
- Cleaning and preprocessing data
- Performing data analysis using appropriate tools
- Creating dashboards and reports for visualization

The internship provided a structured environment similar to industry standards, where tasks were completed within deadlines and required systematic execution. It helped in improving technical skills as well as communication and problem-solving abilities.

This experience played an important role in bridging the gap between academic learning and practical application in the field of data analytics.

INTERNSHIP SPECIFICATION

Data Analytics

Data analytics is the process of collecting, cleaning, analyzing, and interpreting data to extract useful insights and support decision-making. It helps organizations understand patterns, trends, and relationships within data.

In this internship, data analytics was applied to analyze food ordering data and understand customer preferences. The project involved working with datasets containing information about food categories, customer choices, and ordering behavior.

The steps involved in data analytics include:

- **Data Collection:** Gathering relevant data from available sources
- **Data Cleaning:** Removing errors, handling missing values, and preparing data
- **Data Analysis:** Identifying patterns and relationships within the data
- **Data Visualization:** Representing data using charts and dashboards
- **Insight Generation:** Drawing conclusions to support decision-making

This internship gave me hands-on experience in the **complete data analytics lifecycle**, from SQL querying and Excel exploration to advanced Power BI dashboards.

Tech Stack Used:

The **tech stack** refers to the set of software tools and technologies used to complete the project. For this internship, I used a combination of **database tools**, **data visualization software**, and **version control platforms** to build the project *"Facebook Food Trends Analysis: Understanding Customer Preferences."*

SQL:

SQL was used to work with structured datasets and extract relevant information required for analysis. It helped in filtering data, retrieving specific records, and organizing datasets for further processing. By using queries, meaningful subsets of data were prepared for analysis.

Microsoft Excel:

Microsoft Excel is a spreadsheet tool that is commonly used in data analysis. It was used in the initial stages of the project to **view and understand the raw dataset**, explore column names, identify missing values, and apply basic filters. Excel also made it easy to quickly verify data transformations done in Tableau Prep Builder.

Python:

Python was used for data processing and analysis. It helped in handling datasets efficiently, performing transformations, and identifying patterns within the data. Python made it easier to work with large datasets and perform logical operations required during analysis.

Power BI:

Power BI was used for creating interactive dashboards and visualizations. It helped in representing data insights in a clear and user-friendly manner. Various charts and reports were created to highlight trends and patterns in customer preferences.

Programming Languages:

- **Python:** Used for data manipulation, basic ETL operations, and analytical tasks. Python offers libraries like Pandas, NumPy, and Matplotlib for data analysis and visualization.

Database Tools:

- **MySQL Workbench:** Used to manage structured datasets, run SQL commands, and understand data structure before visualization.
- **CSV / Excel Files:** Used as primary data sources for importing and cleaning data.

Data Visualization Setup:

The data visualization environment was set up using Power BI. The datasets were imported into the tool, and necessary data transformations were performed before creating visualizations. The platform allowed easy integration of datasets and provided features such as filters, calculated fields, and interactive dashboards. This setup enabled efficient analysis and clear representation of insights derived from the data.

INTERNSHIP REQUIREMENTS

The internship project, "*Food Trends Analysis: Understanding Customer Preferences*," focuses on analyzing customer behavior and identifying patterns in food ordering data. The project aims to transform raw data into meaningful insights that can support business decisions such as menu planning, marketing strategies, and customer engagement.

Journey

The project follows a structured workflow similar to real-world data analytics practices:

Dataset Collection: The dataset consists of food-related data, including customer preferences, food categories, order frequency, and ratings. This data forms the foundation for analysis.

Data Understanding: The dataset is explored to understand its structure, attributes, and relationships between different variables.

Data Cleaning & Preparation: The data is processed by handling missing values, removing inconsistencies, and formatting it properly for analysis.

Data Analysis: Analytical techniques are applied to identify trends, patterns, and relationships in customer behavior.

Visualization Creation: Charts and graphs are created to represent insights clearly using visualization tools.

Dashboard Development: Multiple visualizations are combined into interactive dashboards to provide a complete view of insights.

Output & Reporting: The final dashboards and findings are documented and presented in a structured format.

Functional Requirements

The system should support the following functionalities:

- Ability to process and analyze food-related datasets
- Identify customer preferences and ordering trends
- Generate visual representations such as charts and dashboards
- Provide insights into popular food categories and patterns
- Allow easy interpretation of data through interactive dashboards

Non-Functional Requirements

The system should meet the following quality requirements:

- Accuracy: Data analysis should provide correct and reliable insights
- Performance: Data processing and visualization should be efficient
- Usability: Dashboards should be simple and easy to understand
- Scalability: The system should handle larger datasets if required

- Reliability: The system should consistently produce stable results

Data Flow

1. **Data Source:**Collection of food-related datasets containing customer preferences and order details
2. **Data Processing:**Cleaning and preparing data by handling missing values and inconsistencies
3. **Data Analysis:**Applying analytical techniques to extract patterns and trends
4. **Visualization:**Creating charts and dashboards to represent insights
5. **Output:**Presenting the final results in the form of dashboards and reports Technology Stack

Technology Stack:

The project uses a combination of tools and technologies for data analysis and visualization:

SQL for data extraction and querying

Python for data processing and analysis

Microsoft Excel for initial data exploration

Power BI for visualization and dashboard creation

These tools work together to complete the end-to-end data analytics process.

Frontend & Visualization:

The visualization layer of the project is handled using Power BI. It provides an interactive interface to represent data insights in a clear and meaningful way.

- Dashboards include charts such as bar graphs, pie charts, and trend analysis visuals
- Filters and slicers are used to improve interactivity
- Data is presented in a structured format for easy understanding

Database & Data Storage:

- **CSV Files (Kaggle):** Source datasets structured with features like food type, food brand, likes,shares and comments. CSV files are easy to import into Powerbi for analysis.

Data Processing & Analysis:

- **Power BI Built-in Tools:** Handle data cleaning (removing nulls, renaming columns) and calculated fields. Charts and KPI cards can be created without external tools, ensuring efficient preprocessing and visualization.

MODULES LEARNED

1. Introduction to Data Analytics and Project Orientation

This initial module gave an overview of the **internship structure**, the **importance of data analytics in real-world projects**, and the role of visualization in decision-making. I understood how data analytics can be applied to the housing market to study patterns like pricing trends and key influencing features.

- Basics of data analytics and visualization
- Understanding project goals and objectives
- Introduction to Powerbi environment and MySQL

2. SQL and Database Management Module

In this module, I learned to work with structured datasets using MySQL Workbench.

- Practiced SQL commands such as SELECT, WHERE, ORDER BY, GROUP BY, and basic joins.
- Explored datasets to understand structure, relationships, and filtering.
- Gained confidence in querying data before visualization.

3. Data Cleaning and Transformation Module

This module focused on preparing and cleaning data before visualization. Using Tableau Prep Builder and Microsoft Excel, I learned how to:

- Handle null and duplicate values
- Rename columns and format data
- Create clean, structured datasets suitable for analysis

  This step helped ensure that the visualizations created later were accurate and meaningful.

4. Data Visualization Using Power BI

In this core module, I learned to build interactive visualizations using Power BI.

- Created different chart types like bar charts, pie charts, histograms, maps, and KPI cards.
- Designed dashboards to display food trends visually and clearly.

5. Dashboard and Storytelling Module

This module helped me combine multiple visualizations into **interactive dashboards** and **stories**.

- Created storyboards to present the step-by-step analysis of the housing dataset.
- Learned how to make visual insights easy to understand for non-technical audiences.

6. Ideation and Documentation Module

Apart from technical learning, I also worked on project ideation and documentation to present the analysis clearly.

- Prepared empathy map, problem statement, and brainstorming templates.
- Wrote project documentation and maintained task updates.

ANALYSIS OF THE MODULES COMPLETED AS A PART OF INTERNSHIP

Introduction to Data Analytics and Project Orientation (Sessions 1–2)

- **Session 1:** Overview of the internship structure, goals, and introduction to the project *"Food Trends Analysis:Understanding Customer Preferences."*
- **Session 2:** Discussed fundamentals of data analytics, importance of visualization in decision-making, and tools to be used such as Power BI and MySQL Workbench.

Analysis:

This module laid the foundation for the entire internship. It provided a clear understanding of the **project objectives**, the **data analytics lifecycle**, and how different tools would be integrated. Understanding the end goal early helped in following a structured approach throughout the internship. It also strengthened the conceptual understanding of why visualization is a powerful tool in data-driven industries.

SQL and Database Management (Sessions 3–5)

- **Session 3:** Installation and setup of MySQL Workbench, database configuration.
- **Session 4:** Practiced SQL commands such as SELECT, WHERE, ORDER BY, and GROUP BY to retrieve and filter structured data.
- **Session 5:** Explored dataset structure, relationships, and basic joins.

Analysis:

This module focused on **data extraction and preparation**. By learning SQL, I was able to **interact directly with structured data**, filter important attributes, and prepare datasets for visualization. SQL also helped build logical thinking and an analytical mindset for handling data. Challenges included understanding complex queries initially, but consistent practice made it easier to structure and retrieve data accurately.

Data Cleaning and Transformation (Sessions 6–8)

- **Session 6:** Imported raw datasets into Power BI.
- **Session 7:** Handled null values, removed duplicates, and renamed fields for consistency.
- **Session 8:** Integrated Microsoft Excel to verify dataset accuracy and formatting.

Analysis:

This module taught the **importance of clean and structured data** in analytics. I learned to identify errors in data, fix inconsistencies, and prepare a dataset ready for visualization. Tableau Prep Builder simplified cleaning tasks, and Excel gave better clarity in spotting issues. This phase was crucial because accurate analysis depends on high-quality data.

Data Visualization Using Power BI (Sessions 9–12)

- **Session 9:** Introduction to chart creation using Tableau Public.
- **Session 10:** Created bar charts, pie charts, line graphs, histograms, and KPI cards.
- **Session 11:** Applied filters, parameters, and calculated fields to make visuals dynamic.

- **Session 12:** Explored maps and geospatial visualizations for food trends.

Analysis:

This module gave me **hands-on visualization experience**. I learned how to convert raw numbers into interactive visuals that clearly communicate insights. Power BI made complex patterns easy to identify, improving analytical thinking. The biggest learning was how visualization can turn raw housing data into **actionable insights**. This was the most creative and impactful part of the internship.

Dashboard Creation and Storytelling (Sessions 13–15)

- **Session 13:** Linked multiple visualizations into dashboards.
- **Session 14:** Used filters, legends, and actions to make dashboards interactive.
- **Session 15:** Created stories to explain analysis step by step.

Analysis:

This module strengthened my ability to **communicate insights effectively**. Instead of just creating visuals, I learned how to arrange and structure them logically in dashboards. Stories made the project presentation easy to follow, which is crucial in real-world analytics where decision-makers prefer visuals over technical reports.

Ideation and Documentation (Sessions 16–18)

- **Session 16:** Created empathy maps, problem statements, and brainstorming templates.
- **Session 17:** Prepared project documentation for dashboards and datasets.
- **Session 18:** Coordinated with team members for structured reporting.

Analysis:

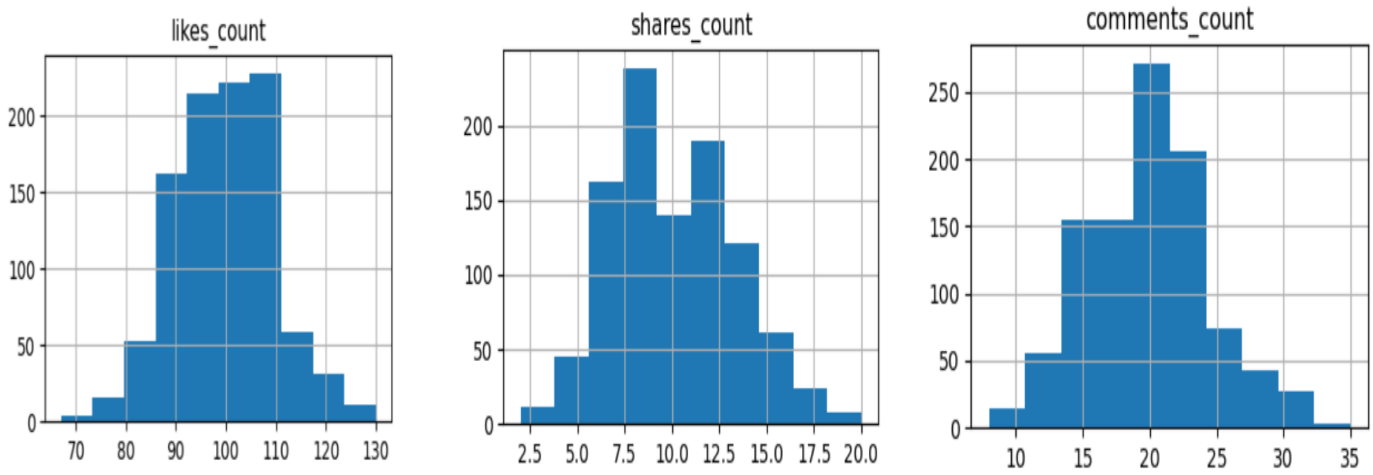
This module emphasized **project planning, communication, and organization**. I learned how to structure ideas clearly before implementation, which improved the quality of my dashboards. Documentation ensured every step of the project was properly recorded, making final submission smooth and professional.

- **Session 19:** Final review and correction of visualizations and dashboards.
- **Session 20:** Compiled all documentation and created the final presentation.
- **Session 21:** Uploaded the completed project to GitHub for final submission.

PROJECT OUTPUT AND VISUALIZATIONS

Exploratory Data Analysis:

1)



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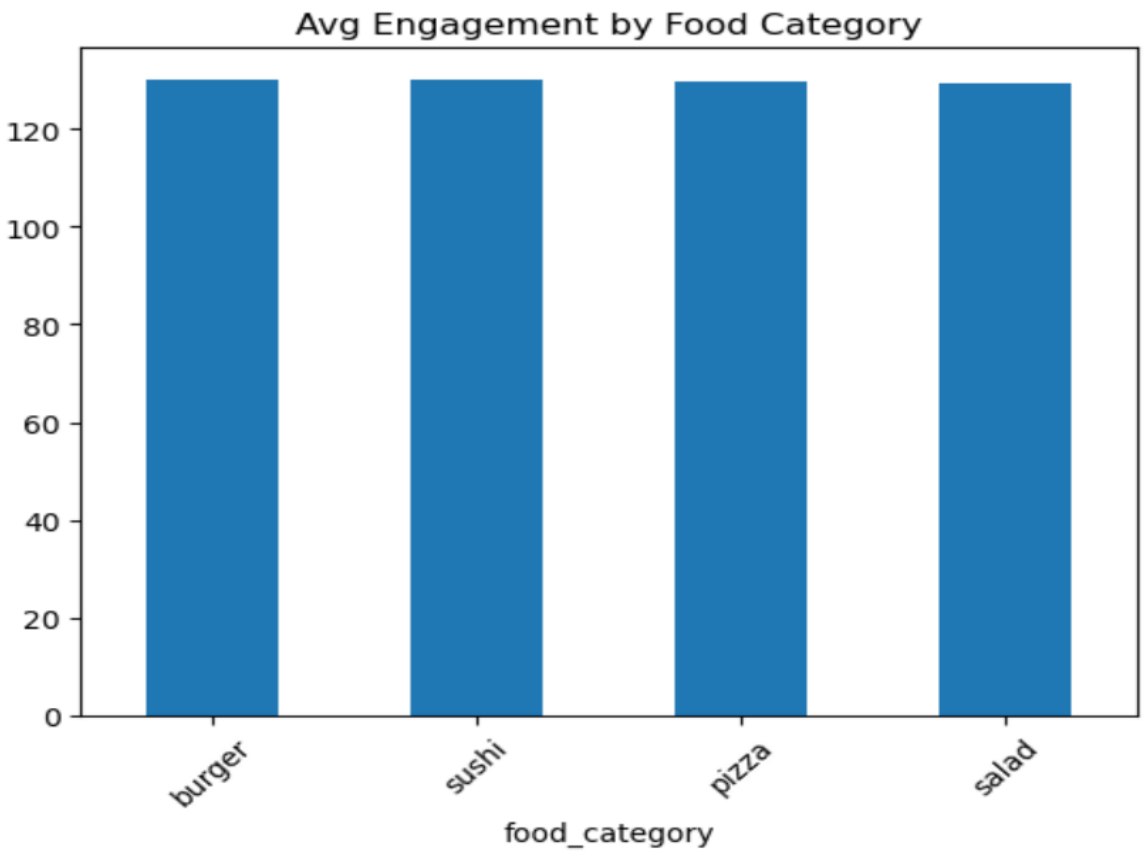

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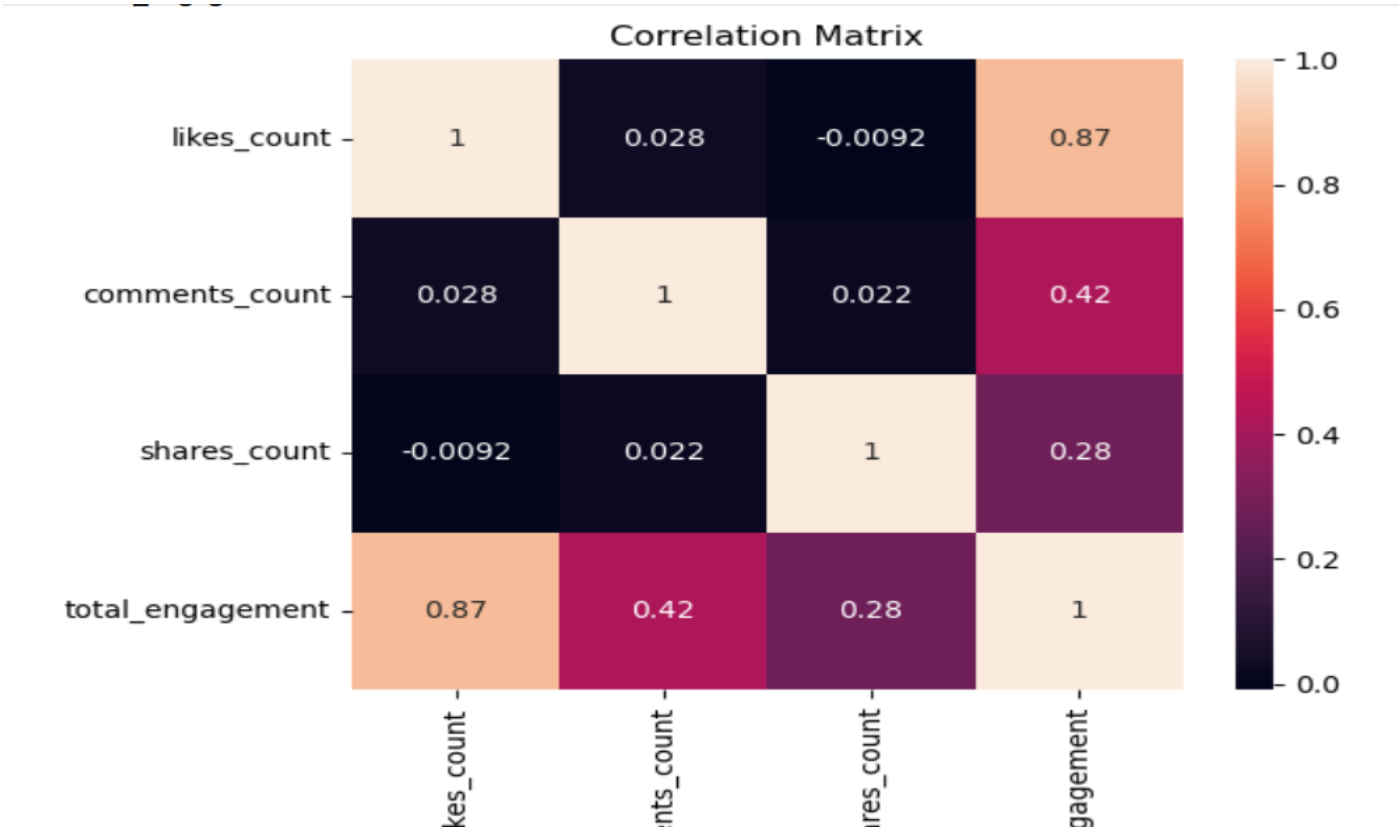
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shares_count
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0.42
-0.0092
0.022
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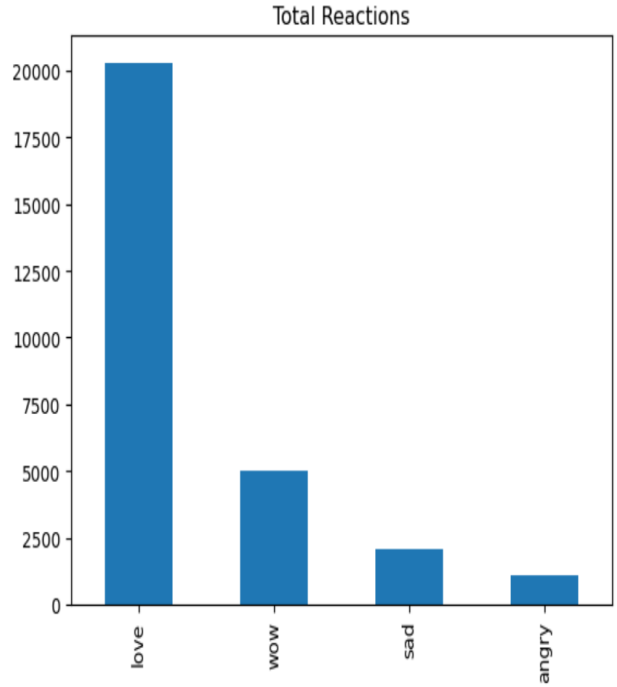
0.8
0.6
0.4
0.2
0.0

4)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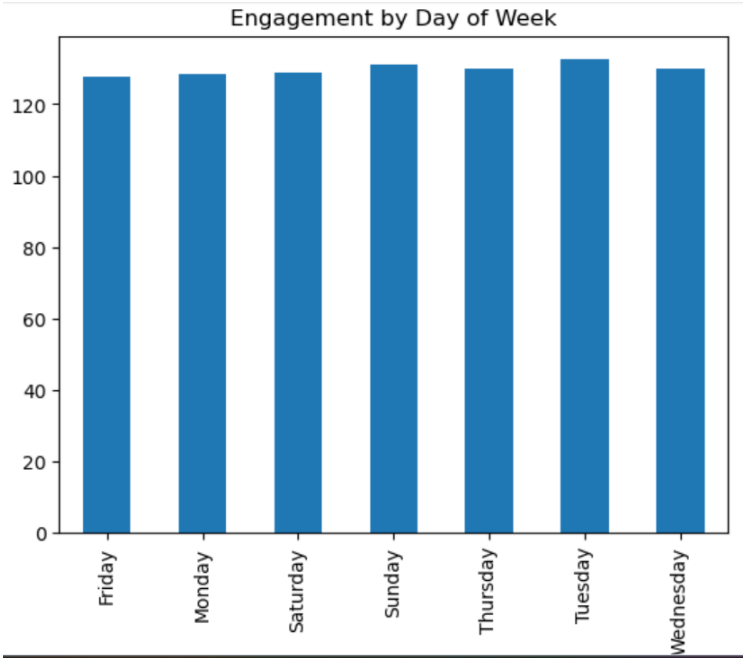
sad
angry


Engagement by Day of Week
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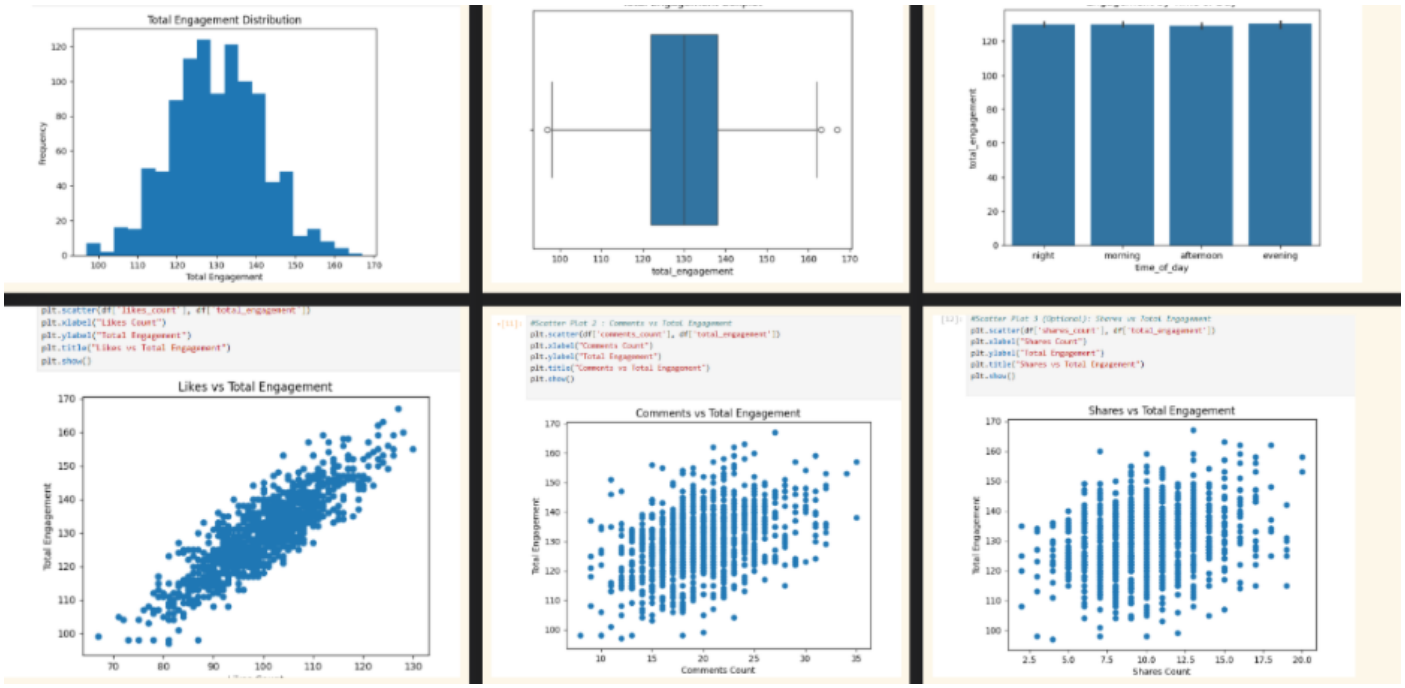
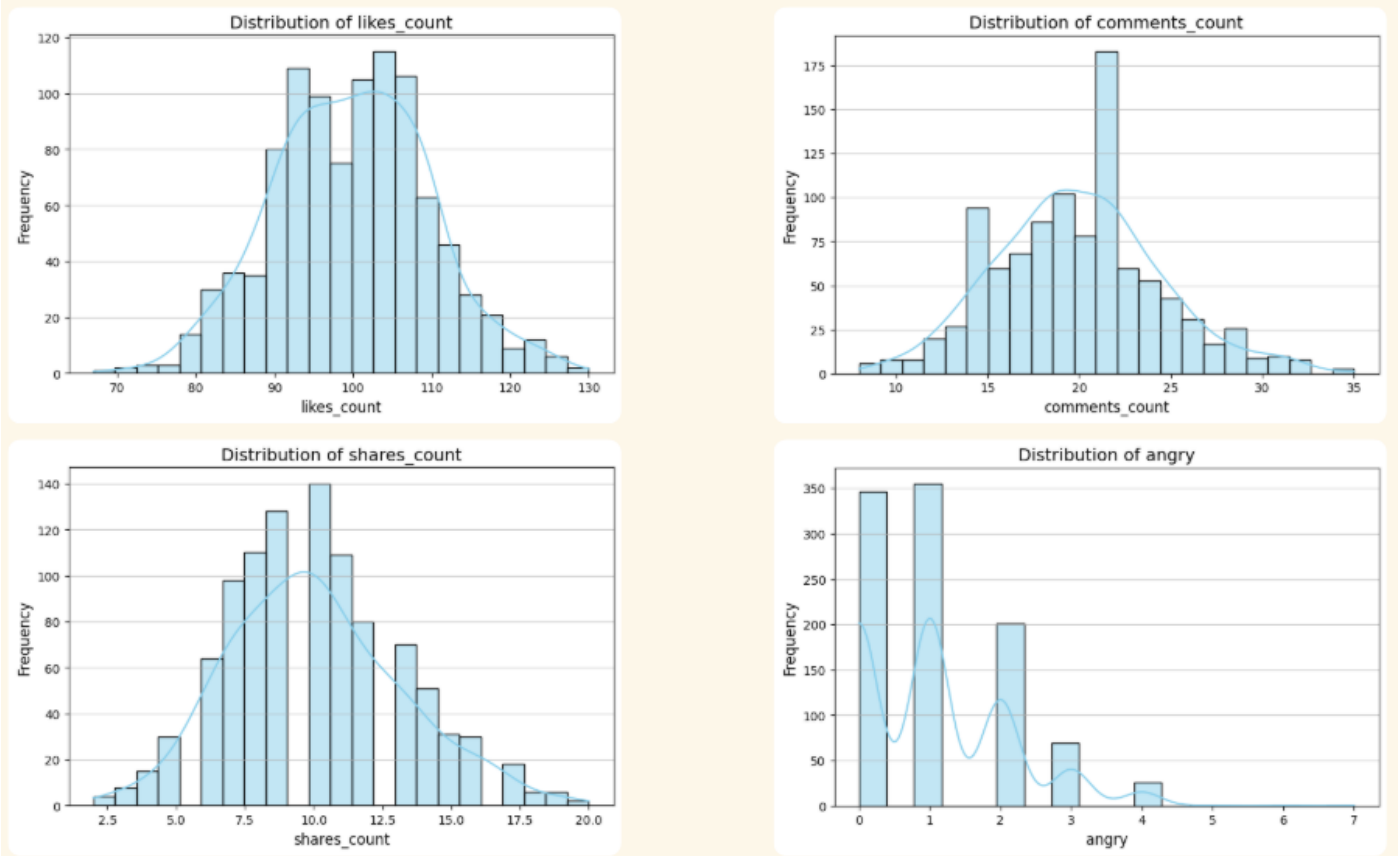
100
80
60
40
20
0
Friday
Monday
Saturday
Sunday
Thursday
Tuesday
Wednesday

visualizations

1)



2)



Dashboard:

Facebook Food Trends Engagement Overview
1000
Count of post_id
100K
Sum of likes_count
10K
Sum of shares_count
130K
Sum of total_engagement
20K
Sum of comments_count
Engagement by Food Category
Engagement Trend Over Time
Engagement by Post Type
video
image
63553
(48.96%)
66244
(51.04%)
Brand Engagement Share
dominos
mcdonalds
subway
chipotle
34K
33K
31K
31K
Audience Reaction Insights
20K
Sum of love
5004
Sum of wow
2094
Sum of sad
1086
Sum of angry
food_category
image
video
Total
sushi
17439
16222
33661
salad
12072
15181
27253
pizza
16142
18531
34673
burger
17900
16310
34210
Total
63553
66244
129797
Reaction Distribution
Sum of love
Sum of angry
Sum of wow
Sum of sad
Engagement by Day of Week
Reactions by Food Category
Content Performance Analysis
99.81
Average of likes_count
19.95
Average of comments_count
10.04
Average of shares_count
129.80
Average of total_engagement
Interaction Type by Food Category
Likes vs Total Engagement
Engagement By Post Type
Posting Strategy Dashboard
129.80
Average of total_engagement
167
Max of total_engagement
4
Count of food_category
4
Count of brand_name
Angry
1086
0
2172
Love
20K
0K
41K
Food Category Share
burger
salad
sushi
pizza
Brands by Engagement
day_of_week
afternoon
evening
morning
night
Total
Friday
124.77
128.00
129.10
128.33
127.76
Monday
129.83
131.75
127.14
126.77
128.35
Saturday
128.23
129.46
128.05
129.19
128.70
Sunday
130.50
130.50
132.02
131.48
131.27
Thursday
127.50
127.33
131.45
131.52
129.97
Tuesday
133.28
132.35
131.06
133.51
132.53
Wednesday
129.03
132.00
130.60
129.56
130.16
Total
129.00
130.15
129.91
130.02
129.80

CONCLUSION

The project "Food Trends Analysis: Understanding Customer Preferences" demonstrates how raw food-related data can be transformed into meaningful insights using data analytics techniques. By analyzing customer behavior, ordering patterns, and food preferences, the project provides a clear understanding of trends that can support business decision-making.

Through the use of tools such as SQL, Python, Microsoft Excel, and Power BI, the data was processed, analyzed, and visualized effectively. The development of interactive dashboards helped in presenting insights in a simple and user-friendly manner, making it easier to identify popular food categories and customer preferences.

This internship at Digit IT, Hyderabad provided valuable hands-on experience in the complete data analytics workflow, including data preprocessing, analysis, and visualization. It helped in improving technical skills, analytical thinking, and the ability to interpret data effectively.

Overall, this experience has strengthened my understanding of real-world data analytics applications and prepared me for future opportunities in the field of data analytics and business intelligence.

INTERNSHIP CERTIFCATE

CERTIFICATE OF INTERNSHIP COMPLETION

To Whom It May Concern

This is to certify that **Boreddy Sravanthi** has successfully completed an internship as a **DATA ANALYST INTERN** at DIGIT IT from January 19th, 2026 to April 29th, 2026.

During the tenure of the internship, she was involved in various tasks and projects related to real-time data analysis. She demonstrated good commitment, professionalism, and a willingness to learn. Her performance and contributions during this period were appreciated.

We appreciate her dedication and wish her success in all future endeavors.

Date: 22/04/2026
Place: Hyderabad

DIGIT IT HYDERABAD

Authorized Signatory
MG Krishna
Manager
DIGIT IT

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